

Avneet Aurora

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EDUCATION

University of Delaware, Newark, DE **B.S. Computer Engineering – Honors College** | GPA: 3.77 | **Dean's List**
Coursework: Calculus III, Linear Algebra II, Data Structures & Algorithms, Software Engineering, Machine Learning, Digital Logic Design

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, Java, HTML, CSS, VHDL

ML & Data: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib

Backend & Cloud: FastAPI, Node.js, REST APIs, SQLite, PostgreSQL, AWS (EC2, RDS), CI/CD, Linux/Bash, Git

Frontend & Mobile: React, Next.js, React Native (Expo)

RELATED EXPERIENCE

Undergraduate Research Assistant — Vertically Integrated Projects (Scientific Computing)

Feb 2026 – Present

- Extended peer-reviewed 2D Lattice Boltzmann fluid-simulation research to 3D (D3Q27 lattice) by replacing the analytical BGK collision operator with a learned neural network, enabling data-driven simulation of turbulent flows
- Improved model generalization and shrank the parameter search space by engineering TensorFlow/NumPy training pipelines that enforce physical lattice-symmetry constraints on the learned operator
- Achieved mass-conservation error at machine precision ($\sim 1e-15$) by validating the 3D model against analytical solutions, confirming the operator preserves the underlying physics; presented results to the research team

Undergraduate Research Assistant — Winter Fellows, Electrical & Computer Engineering

Jul 2025 – Feb 2026

- Raised wildfire-ignition validation accuracy by 10–20% over a baseline stuck at $\sim 50\%$ by re-architecting a PyTorch CNN (ReLU, batch normalization, regression head), correcting the data pipeline, and expanding the training set
- Selected for the merit-based Winter Fellows award to extend 3D ignition-estimation research from airborne to satellite LiDAR point clouds; presented findings at the Winter Undergraduate Research Showcase
- Reduced per-plot preprocessing from a manual multi-step workflow to a single scripted run by automating Python pipelines for point-cloud normalization, forest-metric computation, and heatmap visualization

Technology Intern — Code Differently – Buccini Pollin Group, Wilmington, DE

Jun 2024 – Aug 2024

- Performed financial analysis for real estate development projects; applied problem-solving and time management skills to present a data-driven proposal to BPG founders on amenity gym improvements.
- Earned Microsoft Office, Excel, Word, and Digital Literacy certifications; awarded a \$1,000 Metropolitan Urban League scholarship

PROJECT EXPERIENCE

Decora AI — [decora.club](#)

Aug 2025 – Nov 2025

- Shipped a production full-stack AI interior-design platform serving live users by building a React/Next.js + TypeScript frontend against RESTful serverless endpoints deployed on Vercel
- Enabled monetization with zero payment-data liability by integrating a Stripe-based credit system with Supabase Auth, delegating card handling to Stripe rather than storing credentials
- Kept user redesigns persistent across sessions by modeling furniture-detection output in Postgres and storing images in Supabase buckets

AI_GroceryShopper — https://github.com/Avi702/AI_GroceryShopper

June 2026 – Jul 2026

- Cut hallucinated item detections by adding a second vision pass that re-checks low-confidence results, capped at 3 retries to limit API cost
- Stopped fabricated product links by pulling URLs from the Google Custom Search API instead of the model, leaving it to format only verified results
- Let the phone app trigger scans from any network without router configuration by having the Raspberry Pi poll the FastAPI backend rather than accept inbound connections
- Lowered per-scan latency and token cost by downscaling grocery captures and routing formatting tasks to a smaller model while reserving vision inference for the larger one

SquatBuddy — [devpost.com/squatbuddy](#) *Mar 2026*

- Delivered joint-specific biomechanical feedback on raw video by training an LSTM autoencoder to detect squat-form anomalies frame-by-frame at HenHacks 2026
- Enabled real-time analysis across the full kinetic chain by engineering an OpenCV video-processing pipeline that extracts and normalizes pose keypoints per frame

LEADERSHIP & AWARDS

DevLab Club — President, Newark, DE

Feb 2025 – Present

- Grew club to 20+ active members by running weekly sessions on full-stack development, Git, REST APIs, and object-oriented design; mentor members on project architecture and debugging

Winter Fellows Award, Dec 2025

Best Technical Implementation Award, Dec 2025 – <https://devpost.com/software/taskai-vjx8cl>